

A Composite Directional Support Gate

Monotone Certificate Fusion with an Outward-Admission Filter

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Abstract

We combine four independently derived lower certificates for the relative fourth singular mode $q_4 = s_4/s_1$: direct Weyl, spectral four-volume with a capacity upper, tail-energy trace volume, and the positive-tail directional packet block. The identity $q_4 = \nu_4/\Lambda_{23}$ makes the composition exact, and taking the maximum preserves validity. The 360-record audit has zero admitted dominance failures. At threshold 10^{-8} , the packet-block route raises full-chain coverage from 113 to 114 of 120 updates. An exact-directional diagnostic suggests 115, but is excluded because one weak record lacks an outward guard between two binary64 Gram assembly paths. The paper therefore adds both a positive composite gain and a precise numerical admission rule.

1 Factorization of fourth-mode support

For nonzero K , let

$$q_j = \frac{s_j(K)}{s_1(K)}, \quad \nu_4 = q_2 q_3 q_4, \quad \Lambda_{23} = q_2 q_3.$$

Then

$$q_4 = \frac{\nu_4}{\Lambda_{23}}. \tag{1}$$

RH-110 enclosed Λ_{23} from recent singular values. RH-111 produced a trace-based lower for ν_4 , while RH-114 produced a PSD directional lower. These estimates can be fused only after making their inequality directions and operator instance explicit.

Theorem 1 (Factorized capacity gate). *Suppose $0 \leq V^- \leq \nu_4(K)$ and $\Lambda_{23}(K) \leq L^+ < \infty$. Then*

$$q_4(K) \geq V^-/L^+.$$

Proof. Use (1) and divide the numerator lower by the denominator upper. All quantities are nonnegative. $\square$

2 Monotone composite gate

Let D^- be a direct lower for q_4 . Let V_i^- be any family of valid normalized four-volume lowers sharing one capacity upper L^+ . Define

$$B_{\text{comp}} = \max \left\{ D^-, \frac{V_1^-}{L^+}, \dots, \frac{V_m^-}{L^+} \right\}. \quad (2)$$

Theorem 2 (Monotone composite gate). $B_{\text{comp}} \leq q_4(K)$. Hence $B_{\text{comp}} \geq \tau$ is a rigorous support certificate at threshold τ .

Proof. The direct term is valid by assumption. Every factorized term is valid by Theorem 1; the maximum of lower bounds remains a lower bound. $\square$

The admitted RH-115 family consists of

$$D_{\text{Weyl}}^-, \quad V_{\text{spectral}}^-/L^+, \quad V_{\text{trace}}^-/L^+, \quad V_{\text{PSD}}^-/L^+.$$

The last term uses RH-114's positive-tail packet-block Gramian. No averaging or model selection is performed: the gate only takes a maximum.

Proposition 3 (Information-optimal fusion). *If the only known relations among nonnegative scalars are $B_i \leq q_4$, then $\max_i B_i$ is the largest universal lower bound that can be inferred from their reported values.*

Proof. The maximum is valid by Theorem 2. Sharpness follows by choosing an admissible scalar instance with $q_4 = \max_i B_i$; all stated inequalities remain true, so any larger claimed lower fails on that instance. $\square$

Thus an arithmetic average would be weaker, while adding positive correction terms would require a proved dependence between candidates. The composite gate is not intended to guess such dependence. It is the strongest fusion available from the one-sided ledger alone.

3 Outward-admissibility filter

The algebra above is exact only when all inequalities enclose the same operator K . Numerical values computed by different assembly paths are not automatically composable.

Proposition 4 (Outward-admissibility filter). *A candidate may enter (2) only if its source operator is the same enclosed K , or if a validated perturbation radius transports its bound to K . Agreement to displayed floating-point digits is not a substitute for such an enclosure.*

This is standard interval logic [Moore, 1966]: a point approximation has no inequality direction. In the present chain, RH-110 and RH-114 assemble mathematically equivalent memory Gramians in different binary64 orders. The ambient matrices agree at ordinary precision, but the fourth cross mode is a near-cancellation quantity. One physical record at $\sigma = 0.02$ therefore produces an exact-directional ratio diagnostic exceeding the RH-110 actual ratio by about 6.93×10^{-7} . It appears three times because the same update is labelled

by three thresholds. Proposition 4 excludes this diagnostic until a cross-assembly outward radius is supplied.

The PSD packet-block term does not exhibit this mismatch: after composition it lies below the RH-110 actual ratio on every record.

The filter also clarifies the role of capacity. A volume lower and a capacity upper may be computed by different algorithms, but they must enclose the same K . If their operator representatives differ by a validated radius, Weyl transport can first enlarge L^+ and reduce V^- . Without that radius, division can amplify a tiny fourth-volume discrepancy, precisely as observed in the excluded record.

4 Five-scale composite audit

We join the archived RH-110, RH-111, and RH-114 records by scale, side, threshold, and time. Every admitted candidate is compared with the archived actual q_4 . The audit has five scales, two channels, and 360 labelled records, of which 234 are on the fine chain.

threshold	direct	trace	PSD block	composite	exact diagnostic
10^{-8}	113	113	114	114	115
10^{-6}	109	105	109	109	109
10^{-4}	98	95	98	98	98

Table 1: Full-chain support counts out of 120. The last column is not admitted into the rigorous composite.

There are zero admitted dominance failures and three diagnostic failures, the latter being the threshold-labelled copies described above. The maximum admitted improvement over direct Weyl is 1.55091×10^{-6} in q_4 . All 78 fine updates per threshold are already certified; the minimum fine composite margin is 6.36133×10^{-4} .

5 Consequence and boundary

The positive conclusion is modest but real: the PSD directional block adds one rigorously certified coarse record. The negative conclusion is equally useful: exact-looking values from separately rounded near-cancellation chains cannot be mixed without an outward transport bound. RH-116 will avoid this interface by optimizing memory depth inside one assembly path.

This paper proves no all-level capacity or directional-tail law, no uniform Stage A, no Hilbert–Polya operator, no zeta-zero identification, and no Riemann Hypothesis result.

References

Ramon E. Moore. *Interval Analysis*. Prentice-Hall, 1966.

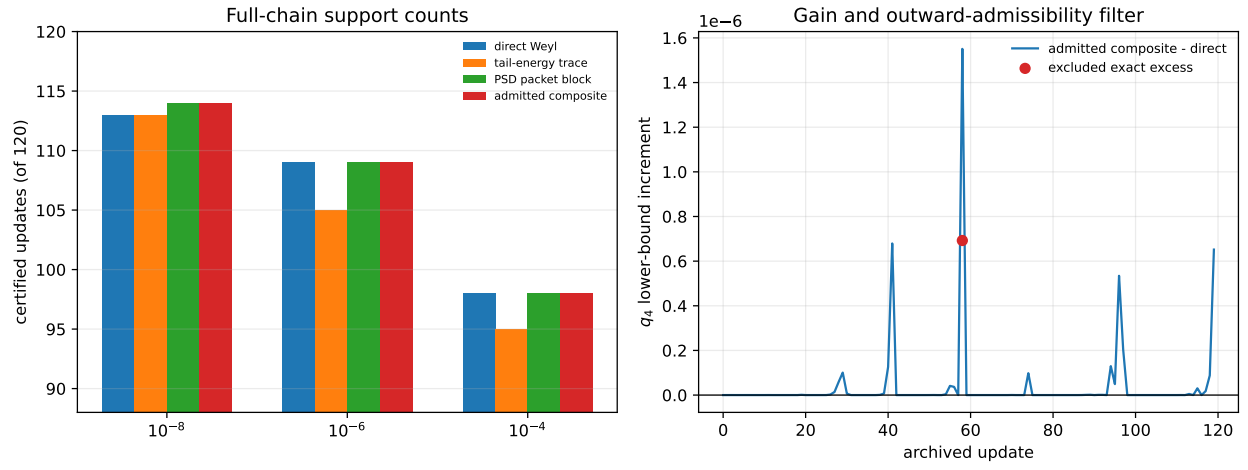


Figure 1: Left: full support counts. Right: admitted gain over direct Weyl and the excluded exact-directional excess.